Assignment No: 1

Title: Design a basic webpage using HTML to demonstrate headings, lists, and images.

Aim: To design a basic webpage using HTML to demonstrate headings, lists, and images.

Objectives:

1. To learn the basic structure of an HTML document.
2. To understand and use different HTML heading tags.
3. To create ordered and unordered lists.
4. To insert and display images using the <img> tag.
5. To understand HTML attributes such as src, alt, width, and height.

Hardware Used:

- Computer or Laptop
- Keyboard
- Mouse
- Minimum 4 GB RAM

Software Used:

- Visual Studio Code (VS Code)
- Web Browser (Google Chrome, Microsoft Edge, or Firefox)
- Operating System: Windows / Linux / macOS

Theory:

1. Introduction to HTML

HyperText Markup Language (HTML) is the standard markup language used to create the structure of webpages. It is used to display text, headings, lists, images, links, and other content in a web browser.

HTML uses tags to define the different elements of a webpage. It is not a programming language; it is a markup language.

2. Basic Structure of an HTML Document

Every standard HTML5 document follows a basic structure consisting of the following elements:

- <!DOCTYPE html>: Declares that the document uses HTML5.
- <html>: The root element that contains the complete HTML document.
- <head>: Contains metadata, page title and character encoding.
- <title>: Specifies the title displayed on the browser tab.
- <body>: Contains all the visible content of the webpage.

3. HTML Headings

HTML provides six heading tags, from <h1> to <h6>. <h1> is generally used for the main title of the webpage, while the other headings are used for subheadings and sections.

Tag	Description
<h1>	Main or most important heading
<h2>	Second-level heading
<h3>	Third-level heading
<h4>	Fourth-level heading
<h5>	Fifth-level heading
<h6>	Sixth-level heading

4. HTML Lists

Lists are used to display items in an organized manner. An unordered list (<ul>) displays items using bullet points, while an ordered list (<ol>) displays items using numbers or other sequence markers. Each item is written using the <li> tag.

Example:

```
<ul>
  <li>Modi</li>
  <li>Devanagari</li>
</ul>

<ol>
  <li>Upload the manuscript image.</li>
  <li>Run the AI decode pipeline.</li>
</ol>
```

5. HTML Images

The <img> tag is used to insert images into a webpage. It is an empty element, meaning it does not require a closing tag. Important attributes of the <img> tag are src (path or URL of the image), alt (alternative text used for accessibility), width and height.

Example:

```
<img src="hero.png" alt="Manuscript" width="400" height="250">
```

6. HTML Attributes

Attributes provide additional information about HTML elements. They are written inside the opening tag as name="value" pairs. An image can be loaded from a local file path or from an online image URL.

Course Project Implementation:

Course-Project Mapping: No file of the LipiSutra course project can be submitted for this assignment. The project's only plain HTML file is index.html, and it is merely the Vite mount shell - it contains a and a module <script> tag, with no headings, lists or images (all visible markup is generated at runtime by React). A fresh HTML file is therefore supplied below; it is themed on LipiSutra so that it sits consistently with the rest of the term work.

Requirement	Present in project?	Where

HTML document skeleton	Yes	index.html
Heading tags h1-h6	No (JSX only)	-
Ordered / unordered lists	No	-
<img> with src/alt	In JSX, not HTML	src/components/Navbar.jsx

Program:

File: assignment01_basic.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>LipiSutra - Manuscript Digitization</title>
</head>
<body>

  <!-- 1. HEADINGS h1 to h6 -->
  <h1>LipiSutra</h1>
  <h2>Digitizing Historical Indian Manuscripts</h2>
  <h3>Project Overview</h3>
  <h4>Supported Scripts</h4>
  <h5>Processing Pipeline</h5>
  <h6>Academic Term Work - Web Technologies (102606)</h6>

  <p>
    LipiSutra is a multimodal platform that reads, reconstructs and preserves
    historical Indian manuscripts. The home page lets a user upload an artifact
    image and receive a transcript, a modern translation and a summary.
  </p>

  <!-- 2. IMAGE with src, alt, width, height -->
  <h3>Sample Artifact</h3>
  <img src="hero.png"
       alt="A scanned palm-leaf manuscript written in the Modi script"
       width="420"
       height="260">
  <p><i>Figure 1: A degraded manuscript submitted to the decoder.</i></p>

  <!-- 3. UNORDERED LIST -->
  <h3>Scripts Recognised by the System</h3>
  <ul>
    <li>Modi</li>
    <li>Devanagari</li>
    <li>Brahmi</li>
    <li>Sharada</li>
  </ul>

  <!-- 4. ORDERED LIST -->
  <h3>Steps to Decode a Manuscript</h3>
  <ol>
    <li>Open the LipiSutra web application in the browser.</li>
    <li>Drag and drop a high-resolution image of the document.</li>
    <li>Click the "Decode &amp;amp; Analyze Artifact" button.</li>
    <li>Read the transcript, translation and summary tabs.</li>
    <li>Listen to the narrated output using text-to-speech.</li>
  </ol>

  <!-- 5. NESTED LIST -->
  <h3>Platform Roles</h3>
  <ul>
    <li>Public / Student
      <ol>
        <li>Upload an artifact</li>
```

```
        <li>Read the AI output</li>
      </ol>
    </li>
    <li>Historian
      <ol>
        <li>Review a pending document</li>
        <li>Correct and verify the transcript</li>
      </ol>
    </li>
    <li>Museum Admin</li>
  </ul>

  <hr>
  <p>&copy; 2026 LipiSutra - Department of Computer Engineering, KBTCOE Nashik.</p>

</body>
</html>
```

Output:

- The browser displays six headings rendered in decreasing font size from <h1> to <h6>.
- The manuscript image is shown at 420 x 260 pixels; if the file is missing, the alt text "A scanned palm-leaf manuscript written in the Modi script" is displayed instead.
- The script list appears with bullet markers and the decoding steps appear numbered 1 to 5.
- The nested list shows the ordered sub-steps indented under each role.

Conclusion:

In this assignment, we designed a basic webpage using HTML. We learned the structure of an HTML document, the six heading tags, ordered, unordered and nested lists, and how to insert images using the <img> tag. We also understood HTML attributes such as src, alt, width and height, and verified that the LipiSutra project has no equivalent static page because its markup is produced at runtime by React.

References:

6. Mozilla Developer Network (MDN) Web Docs - HTML Basics, https://developer.mozilla.org/en-US/docs/Learn_web_development/Core/Structuring_content
7. W3Schools - HTML Tutorial, https://www.w3schools.com/html/
8. HTML Living Standard - WHATWG, https://html.spec.whatwg.org/